Question 1 Report

Roll Number: `b23cs1075`

1. Objective

This report summarizes the Question 1 pipeline for manual cepstral feature extraction, spectral leakage analysis, cepstrum-based voiced/unvoiced segmentation, and phonetic mapping with a Hugging Face CTC model.

2. Dataset and Audio Files

- Dataset used: `openslr/librispeech_asr`
- Hugging Face subset / split: `clean / test`
- Number of clips analyzed: `3`
- Manifest file: `q1/data/librispeech_subset/manifest.csv`
- Sampling rate used: `16 kHz`

The three analyzed clips were:

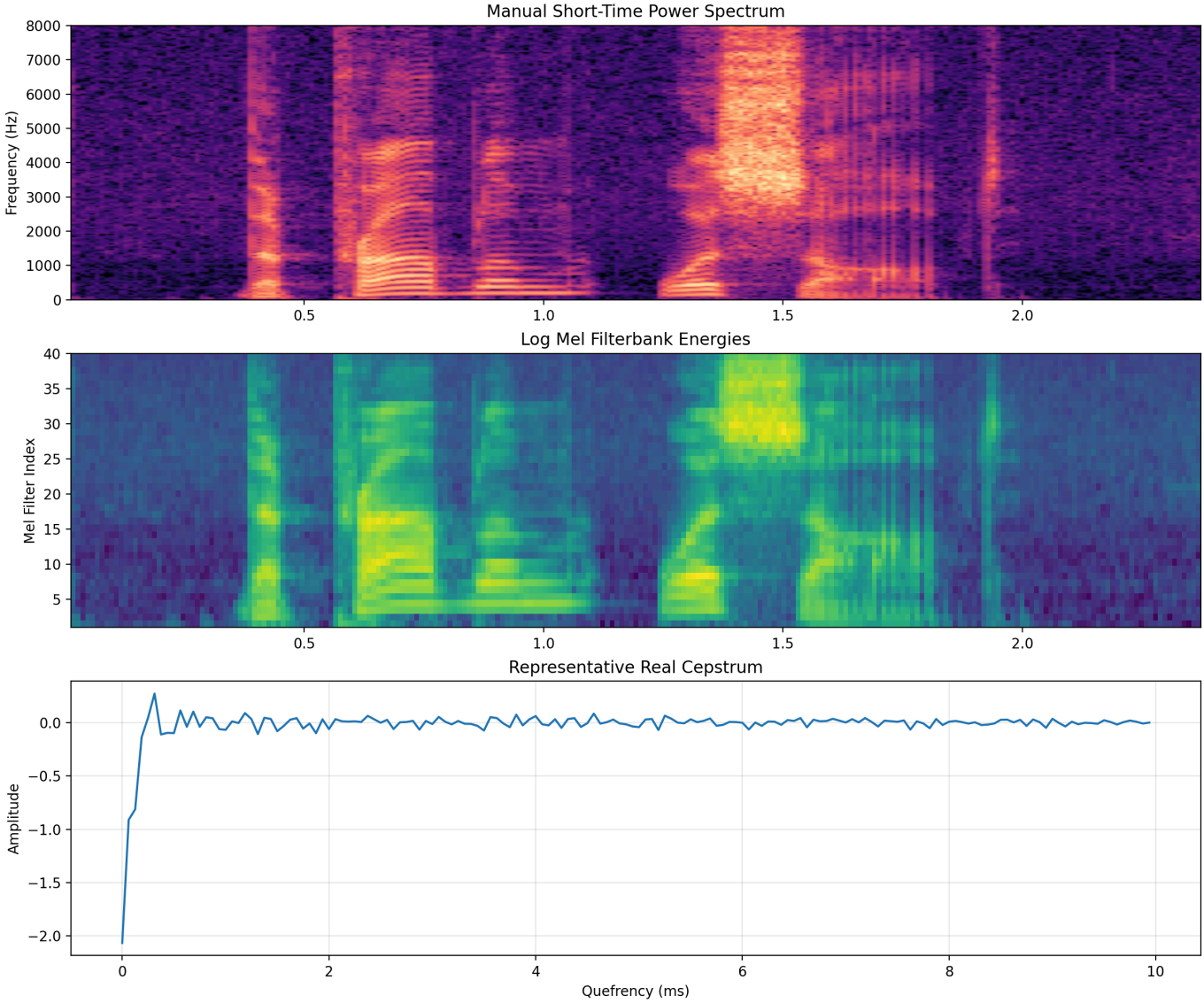
- `2300-131720-0039.wav`: "THE PROBLEM WAS SOLVED"
- `3575-170457-0025.wav`: a long declarative sentence with relatively stable speech continuity
- `4992-41797-0017.wav`: a longer utterance with multiple voiced and unvoiced transitions

3. Manual MFCC / Cepstrum Engine

- Pre-emphasis coefficient: `0.97`
- Frame / hop size: `25 ms / 10 ms`
- Window: `Hamming`
- FFT size: `512`
- Number of Mel filters: `40`
- Number of cepstral coefficients: `13`

The handcrafted pipeline follows the lecture sequence exactly: pre-emphasis, framing, Hamming windowing, real FFT, power spectrum estimation, triangular Mel filterbank projection, log compression, and a manual DCT-II to obtain the cepstral coefficients. In parallel, the real cepstrum is computed from the log spectrum to analyze low- and high-quefrency structure.

The representative MFCC / cepstrum figure was chosen from `2300-131720-0039.wav` because the clip is short and the word transitions are visually easy to interpret. The figure already combines the short-time spectrum, log-Mel energies, and a representative real cepstrum in one panel.



4. Spectral Leakage and SNR Analysis

The selected speech segment was an automatically chosen high-energy region, and the metrics were defined as follows:

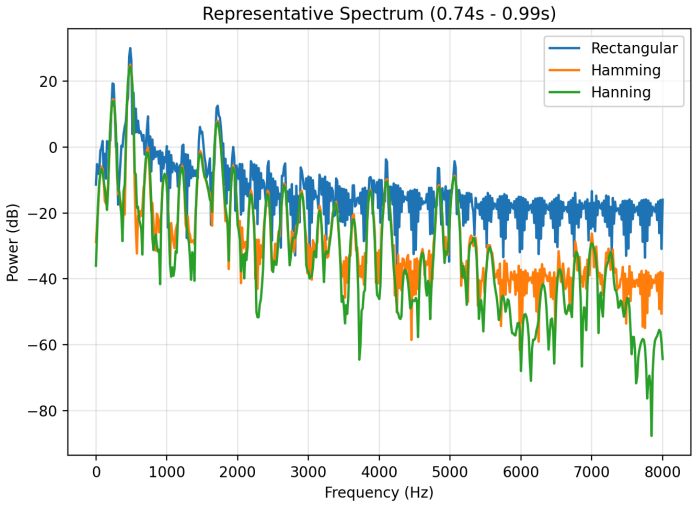
- Leakage percentage: off-main-lobe energy divided by total spectral energy.
- SNR: dominant-peak energy divided by off-main-lobe energy.

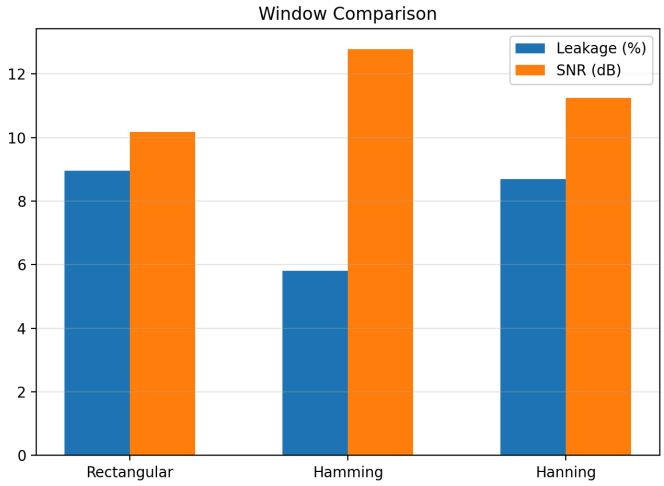
The representative leakage comparison was chosen from `4992-41797-0017.wav` over the automatically selected high-energy segment from `0.74 s` to `0.99 s`. This sample showed the clearest difference between window types.

Window	Leakage (%)	SNR (dB)
Rectangular	8.9536	10.1712
Hamming	5.8041	12.7856
Hanning	8.6907	11.2490

For this representative segment, the Hamming window gave the best trade-off: it produced the lowest leakage and the highest SNR. The Hanning window improved over the rectangular case, but it did not

match Hamming on either metric for this clip. Across the three analyzed files, Hamming also gave the best or near-best SNR consistently, so it is the most defensible default choice for the rest of the pipeline.



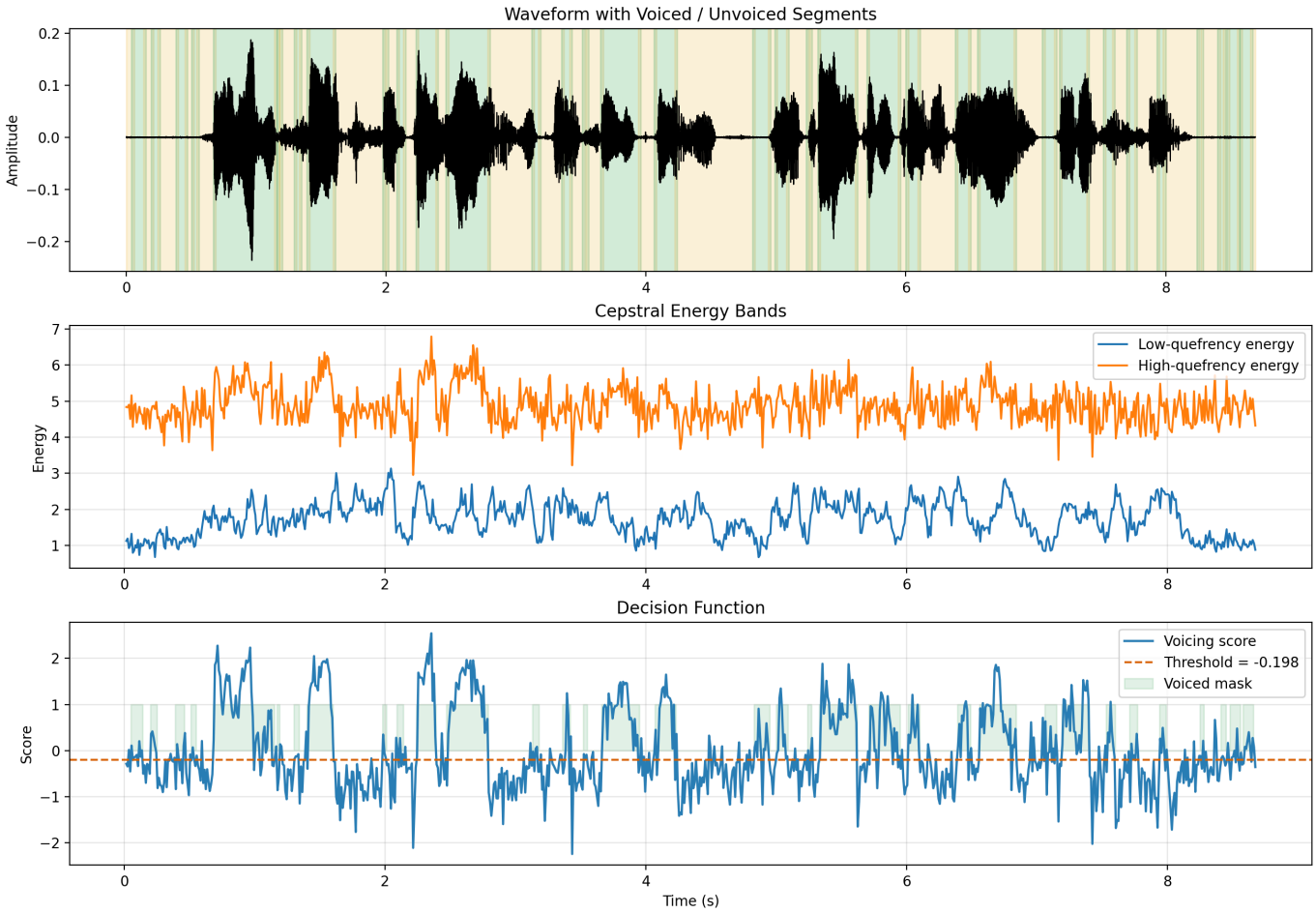


5. Voiced / Unvoiced Boundary Detection

Describe:

- Low-quefrency band used for the vocal-tract envelope: `0.5 to 2.5 ms`
- High-quefrency / pitch band used for periodicity: `2.5 to 16 ms`
- Voicing score and thresholding rule: a z-scored combination of periodicity ratio, high-quefrency energy, and RMS energy, thresholded at the median frame score
- Post-processing or smoothing: median filtering with `5` frames plus short-run cleanup

The voiced / unvoiced visualization was chosen from `4992-41797-0017.wav` because it contains many transitions and therefore shows the segmentation behavior more clearly than the very short clip. For this sample, the detector produced `69` segments with a voiced-frame ratio of `0.4694` and an adaptive threshold of `-0.1978`.



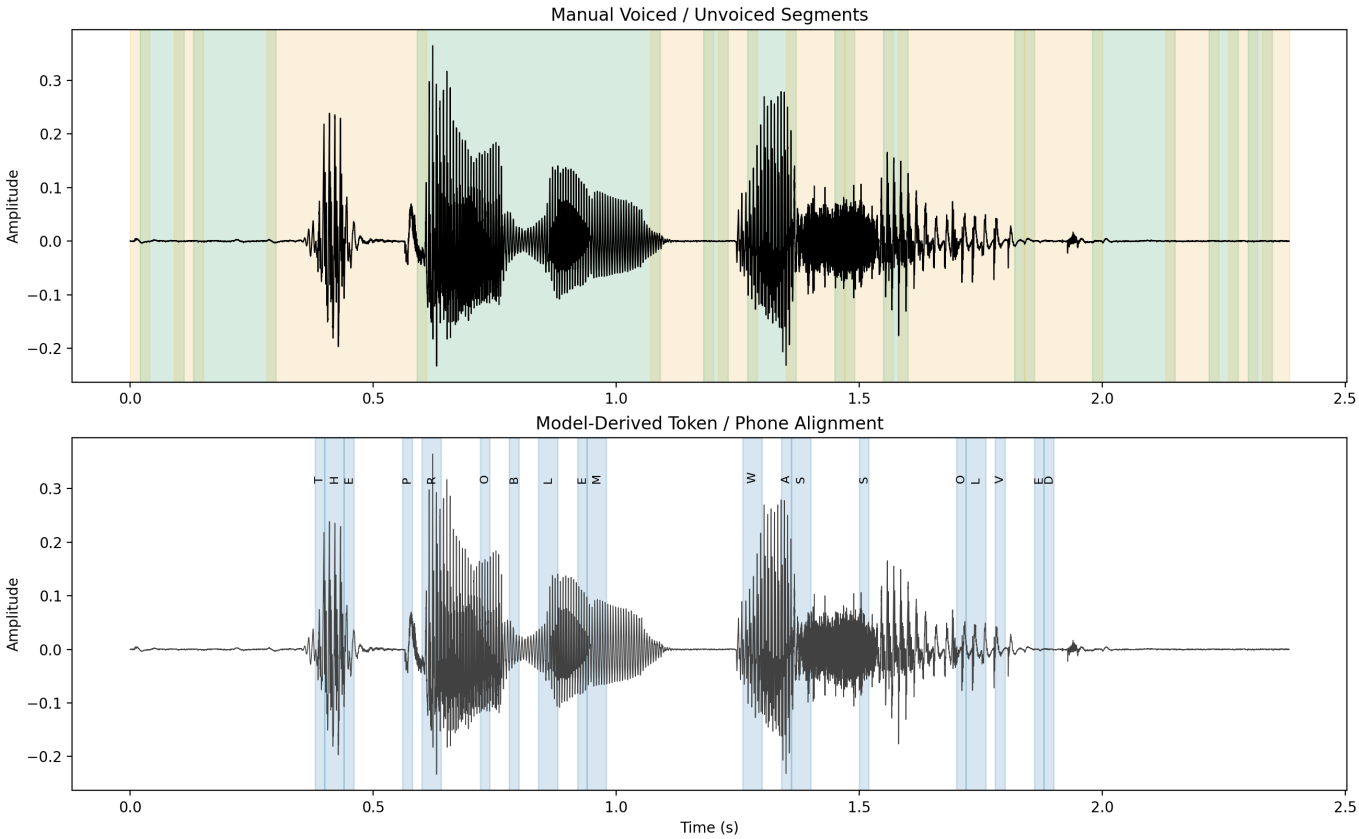
6. Phonetic Mapping and RMSE

- Hugging Face model name: `facebook/wav2vec2-base-960h`
- Alignment mode (`forced` if transcript/phone sequence provided, otherwise `greedy`): `forced`
- Boundary RMSE (seconds): per-clip values shown below, with a mean RMSE of `0.1716 s`

Audio	Manual boundaries	Model boundaries	RMSE (s)
`2300-131720-0039.wav`	22	18	0.2256
`3575-170457-0025.wav`	40	119	0.0302
`4992-41797-0017.wav`	68	90	0.2589

The alignment visualization was chosen from `2300-131720-0039.wav` because the short utterance makes it easier to compare broad manual voiced/unvoiced regions against the much denser token-level alignment produced by the CTC model.

The model alignment agrees best with the manual boundaries on `3575-170457-0025.wav`, where the long continuous utterance yields a very low RMSE. The disagreement is larger on `2300-131720-0039.wav` and `4992-41797-0017.wav`, where the manual segmentation captures broad voicing transitions while the CTC model inserts many more fine-grained token boundaries. This gap is expected because the chosen model is character-token based rather than a true phoneme-aligner, so the comparison is a useful approximation rather than a strict phone-boundary ground truth.



7. Limitations

- The leakage and SNR metrics are heuristic on natural speech and depend on the dominant-peak definition.
- Voiced/unvoiced boundaries are frame-based, so transitions are limited by hop resolution.
- Phone-level alignment quality depends strongly on the selected Hugging Face checkpoint and whether a phone sequence is available.

8. Reproducibility

Commands used:

```
python3 q1/prepare_librispeech_subset.py \
  --subset clean \
  --split test \
  --num-samples 3 \
  --auto-phone-sequence \
  --output-dir q1/data/librispeech_subset

python3 q1/run_q1_pipeline.py \
  --manifest-path q1/data/librispeech_subset/manifest.csv \
  --alignment-source transcript \
  --model-name facebook/wav2vec2-base-960h
```